

DSP — Theory Cheat Sheet

Homework 01, Q1 | Basic Concepts Reference

Signal classification & representation (Q1a–c)

Q1a

Ways to classify a signal

- Continuous-time vs. discrete-time (time axis)
- Analog vs. digital (both axes considered together: continuous amplitude = analog; discrete amplitude + discrete time = digital)
- Periodic vs. aperiodic
- Deterministic vs. random (stochastic)
- Causal vs. non-causal (zero for $t < 0$ vs. defined for all time)
- Energy signal (finite total energy) vs. power signal (finite average power)
- 1-D (time) vs. multi-dimensional (e.g. image = $f(x,y)$)

Q1b

Mathematical vs. physical representation

Mathematical: an idealized formula (e.g. $x(t) = \sin(2\pi t)$) — perfectly precise, can extend to $\pm\infty$, zero noise.

Physical: the actual real-world quantity (voltage, pressure, light). Always limited by noise, finite bandwidth, component tolerances.

Key insight: the mathematical model is an idealization of the physical reality.

Q1c

Continuous-time vs. discrete-time vs. digital signals

Continuous-time: defined for every t ; amplitude and time both continuous.

Mathematically clean; physically only an approximation.

Discrete-time: defined only at integer sample instants n . Amplitude still idealized as perfectly precise.

Digital: both time and amplitude are discrete (quantized). The only type whose mathematical and physical representations can be identical — finite precision is a feature, not a limitation.

Systems (Q1d–f)

Q1d

Concept of a system

A system maps an input signal to an output signal: $y = T\{x\}$.

Mathematically represented as a difference equation (discrete-time) or differential equation (continuous-time).

Example: $y[n] = x[n] - x[n-1]$ computes the difference between consecutive samples (discrete derivative).

Q1e

Continuous-time vs. discrete-time system

Continuous-time system: input and output are both continuous-time signals. Example: analog amplifier or RC low-pass filter.

Discrete-time system: input and output are both discrete-time signals. Example: digital equalizer in an audio app.

Q1f

Is a continuous-time system the same as an analog system?

Disagree (strictly). Analog implies both continuous time AND continuous amplitude.

Continuous-time only constrains the time axis — a continuous-time system with quantized amplitude would not be called analog.

In practice the terms are often used interchangeably, but formally analog is the stronger, more specific term.

Interface systems, ADC & DAC (Q1g–j)

Q1g Why do we need interface systems?	Analog and digital systems speak different 'languages'. Interface systems translate between them. Block diagram of a DSP pipeline: Analog input → [ADC] → Digital processor → [DAC] → Analog output (e.g. microphone) (algorithm) (headphones)
Q1h Analog-to-Digital (A/D) converter	Converts a continuous-time, continuous-amplitude signal into a digital (discrete-time, discrete-amplitude) signal. Two steps: sampling (discretize time at rate f_s) and quantization (discretize amplitude into 2^N levels for N-bit ADC).
Q1i Digital-to-Analog (D/A) converter	Converts a digital signal back into a continuous-time analog signal. Steps: reconstruction (assign a pulse/level to each sample) followed by a smoothing (reconstruction) filter.
Q1j Practical vs. ideal ADC/DAC	Ideal ADC: infinite resolution (no quantization error), infinite bandwidth, zero conversion time. Practical ADC: finite bit depth (quantization noise), limited sampling rate, conversion latency, aperture jitter. Ideal DAC: perfect reconstruction, zero glitches, infinite output bandwidth. Practical DAC: glitches between samples, output filtering needed, finite settling time.

Signal processing: forms & comparison (Q1k–n)

Q1k What is signal processing? Forms used in practice.	Signal processing: modifying or analyzing signals to extract information or improve quality. • Analog signal processing (ASP): filtering, amplification of continuous signals. Example: FM radio tuner. • Digital signal processing (DSP): algorithms on sampled data. Example: MP3 audio compression. • Mixed-signal processing: combines both. Example: a smartphone audio codec.
Q1l Analog Signal Processing (ASP) block diagram	Analog input → [Analog system (filter/amplifier)] → Analog output No ADC or DAC needed. Signal stays in the continuous domain throughout. Example: an active low-pass filter built from op-amps and RC components.
Q1m Digital Signal Processing (DSP) block diagram	Analog in → [Anti-alias filter] → [ADC] → [Digital processor] → [DAC] → [Reconstruction filter] → Analog out The anti-alias filter before the ADC prevents aliasing (must attenuate above $f_s/2$). The reconstruction filter after the DAC smooths the staircase output.

Q1n

Why is DSP preferred over ASP?
Any disadvantages?

Advantages of DSP:

- Flexibility: software can be updated without changing hardware.
- Precision: no component drift over temperature or time.
- Reproducibility: two digital filters behave identically; two analog circuits never do.
- Storage & transmission: digital signals can be copied and stored perfectly.

Disadvantages:

- Quantization error: amplitude information lost permanently at the ADC (finite bit depth).
- Bandwidth limit: ADC cannot keep up with very high-frequency signals (GHz range — radar, 5G).
- Latency: ADC/DAC conversion takes time; problematic for hard real-time systems.
- Power & cost: ADC/DAC hardware adds complexity and energy consumption.